



FPT UNIVERSITY

Final Report in Course DPL301m

Using YOLOv12 And OCR For Detection And Recognition License Plate

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JULY 22, 2025

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Abstract

In this project, we develop an Automatic License Plate Recognition (ALPR) system tailored for static vehicle images, focusing on Vietnamese license plates. Our system consists of two main components: (1) a YOLOv12-based object detection module for accurately localizing license plates, and (2) an OCR module for recognizing the alphanumeric characters on the plates.

To enhance model generalization and robustness, we aggregate and reformat multiple datasets—including Roboflow’s “License Plate Recognition”, “Bien so xe”, and a GitHub dataset—into a unified set of **19,248 labeled images**. The YOLOv12 model trained on this enriched dataset achieves state-of-the-art performance in detection, with **0.986 precision, 0.981 recall, 0.992 mAP@0.5**, and **0.761 mAP50-95**.

The OCR module is trained specifically to handle Vietnamese license plate fonts and layouts. It achieves high character-level accuracy, with **mAP@0.5 reaching 0.985** on the validation set. Together, these modules create a robust and efficient ALPR pipeline for traffic and security applications in Vietnam.

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1 Introduction to Data and Problem Statement

In the context of growing urbanization and the increasing need for efficient traffic management, Automatic License Plate Recognition (ALPR)[1] systems play a pivotal role. ALPR not only aids in traffic monitoring and parking management but also serves as a critical tool in public security and law enforcement. This problem typically involves two main stages: **detecting the license plate** within an image using **Yolov12**[2], and subsequently applying **Optical Character Recognition (OCR)**[3] to extract the characters and numbers from the detected plate.

For the initial development and evaluation of our license plate detection capabilities, we utilize the “**License Plate Recognition**” dataset, sourced from Roboflow Universe Projects (ID: `license-plate-recognition-rxg4e`). This dataset is specifically curated for object detection tasks, comprising a total of **10,125 annotated images**. The dataset is judiciously split into:

- **7,057 images** for the training set
- **2,048 images** for the validation set
- **1,020 images** for the test set

Each image in this dataset is accompanied by precise **bounding box annotations**[7], delineating the exact location of license plates. The dataset focuses on a single object class, “license-plate”, simplifying the detection challenge to concentrate on localization accuracy. The diversity in capture conditions (angles, lighting, environments) within this dataset aids in the initial generalization capabilities of the model.

Our project specifically addresses the ALPR problem for **static image inputs** provided via a web application, rather than real-time video streams. This approach allows users to upload an image containing a vehicle, and our system will then process it to identify and read the license plate. We will employ the advanced **YOLOv12 (You Only Look Once version 12)**[2] model for the license plate detection phase. YOLOv12 is a state-of-the-art convolutional neural network architecture known for its high accuracy and efficiency in object detection.

Initially, the YOLOv12 model will be trained and evaluated exclusively on the aforementioned Roboflow dataset to establish a baseline performance. Following this foundational training, our strategy involves **incorporating additional, diverse datasets** to further optimize the model’s robustness and accuracy across a wider range of scenarios and variations in license plate appearance. This iterative approach of expanding the training data aims to enhance the model’s generalization capabilities, ensuring improved performance in real-world conditions. Once YOLOv12[2] accurately identifies and crops the license plate, we will integrate an **OCR module**[3] to convert the cropped license plate image into a text string, thereby completing the automated license plate recognition process for web-based image submissions.

2 Related Work

Automatic License Plate Recognition (ALPR) has been a significant research area in computer vision for decades, with continuous advancements driven by new deep learning techniques. ALPR systems typically involve two sequential stages: detecting the license plate within an image or video frame, followed by recognizing the characters on the detected plate.

Early approaches to ALPR often relied on traditional image processing techniques such as edge detection, morphological operations, and connected component analysis for plate localization, combined with template matching or classical machine learning for character recognition. However, these methods often struggled with varying lighting conditions, complex backgrounds, and diverse plate designs.

The advent of deep learning has revolutionized ALPR, leading to more robust and accurate systems. Contemporary research largely focuses on leveraging convolutional neural networks (CNNs) for both detection and recognition tasks. For the detection phase, You Only Look Once (YOLO) models have proven to be highly effective due to their speed and accuracy. [4] presented a comprehensive ALPR system that leverages YOLOv2 for license plate detection, integrating it with an OCR module for character recognition. Their work highlights the efficacy of combining deep learning-based detection with a separate recognition component in challenging, unconstrained environments.

Further extending the application of YOLO in ALPR, [5] proposed an efficient and layout-independent ALPR system built upon the YOLO detector. Their method focuses on addressing variations in license plate layout and environmental factors, demonstrating how a robust YOLO-based detection module can be foundational for a versatile ALPR solution. This work underscores the strength of YOLO in handling diverse real-world conditions relevant to license plate localization.

For the recognition phase, methods bypassing traditional OCR by directly recognizing character sequences have gained traction. An example is *LPRNet*, which proposes an end-to-end network for license plate recognition without explicit character segmentation, directly predicting the character sequence using a Connectionist Temporal Classification (CTC) loss [6]. This approach is highly efficient and demonstrates strong performance in real-time scenarios, showcasing an alternative to traditional OCR pipelines for character extraction.

Our project builds upon this rich body of work by utilizing the latest iteration, YOLOv12, for the detection of license plates. This represents a significant advancement over earlier YOLO versions like YOLOv2, offering improved accuracy and efficiency. Following the detection, we integrate a separate OCR module for character recognition, akin to the two-stage approach explored by [4] and [5]. Our methodology initially focuses on a dedicated Roboflow dataset for establishing a baseline, with future plans to incorporate additional diverse datasets to further enhance the model's generalization and robustness, addressing the varied challenges

encountered in real-world ALPR applications.

3 Data Exploration

Understanding the characteristics of our dataset is paramount for building a robust and effective Automatic License Plate Recognition (ALPR) system. This exploratory data analysis (EDA) aims to unveil key insights into the "License Plate Recognition" dataset, providing a foundational understanding of its image properties and the distribution of license plate annotations. By thoroughly examining these aspects, we can make informed decisions regarding preprocessing strategies and model configuration.

Our dataset, sourced from Roboflow Universe Projects, comprises a total of **10,125 annotated images**. This dataset is judiciously split into:

- **7,057 images** for the training set
- **2,048 images** for the validation set
- **1,020 images** for the test set

A preliminary check confirmed the integrity of the dataset splits, verifying that the number of image files perfectly matches the number of corresponding label files (‘.txt’) for each subset: 7,057 images and 7,057 label files for the training set; 2,048 images and 2,048 label files for the validation set; and 1,020 images and 1,020 label files for the test set. This consistency indicates a well-organized dataset ready for further analysis. The dataset contains only one object class: **License_Plate**, with a total of **10,637 bounding boxes** annotated across all images, implying that some images may contain more than one license plate.

3.1 Image Dimensions Analysis

Understanding the dimensions of the images in the dataset is crucial for determining appropriate input sizes for our deep learning model. The statistical summary of image dimensions is presented below:

Image dimension statistics:

	Width	Height
count	10125.000000	10125.000000
mean	577.362963	407.500642
std	400.292456	443.306813
min	90.000000	86.000000
25%	472.000000	303.000000
50%	472.000000	303.000000
75%	472.000000	303.000000
max	5568.000000	4608.000000

The mean image width is approximately 577 pixels, and the mean height is around 407 pixels. Notably, there's a significant standard deviation, indicating a wide range of image sizes, from a minimum of 90x86 pixels to a maximum of 5568x4608 pixels. The 25th, 50th (median), and 75th percentiles for both width and height are identical (472x303), suggesting a high concentration of images around this specific resolution.

The distribution of image widths and heights is visualized in Figure 1.

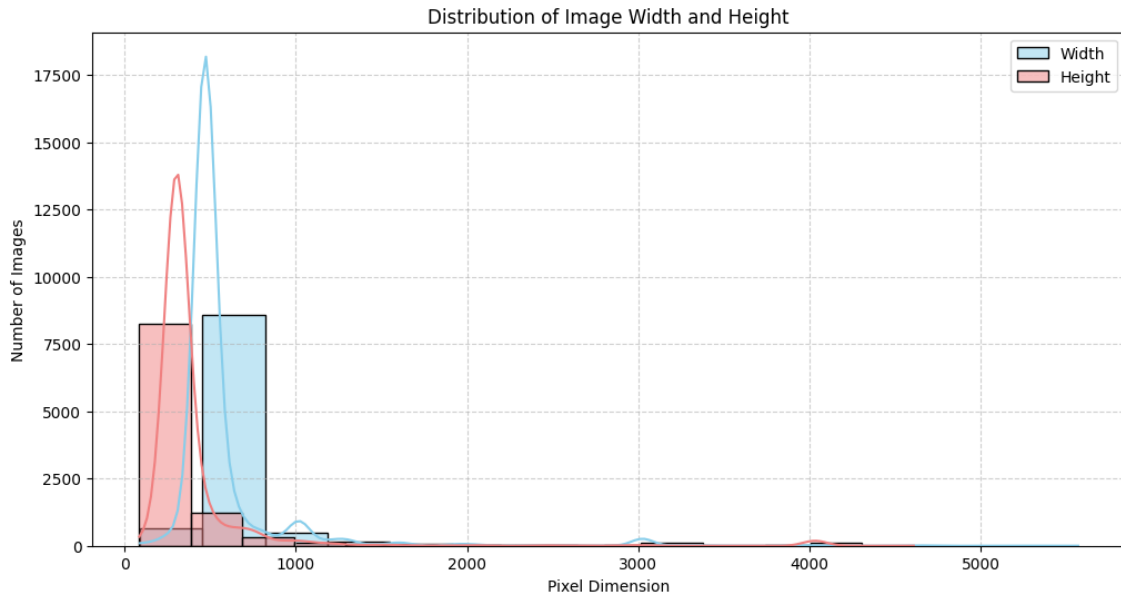


Figure 1: Distribution of Image Width and Height

Figure 1 reinforces the observation from the statistics, showing prominent peaks around 472 pixels for width and 303 pixels for height, confirming that a substantial portion of the dataset comprises images with these dimensions. However, the presence of longer tails and smaller peaks indicates a scattering of images with much larger dimensions, necessitating a robust resizing strategy for model input.

Further insight into the relationship between image width and height is provided by the scatter plot in Figure 2.

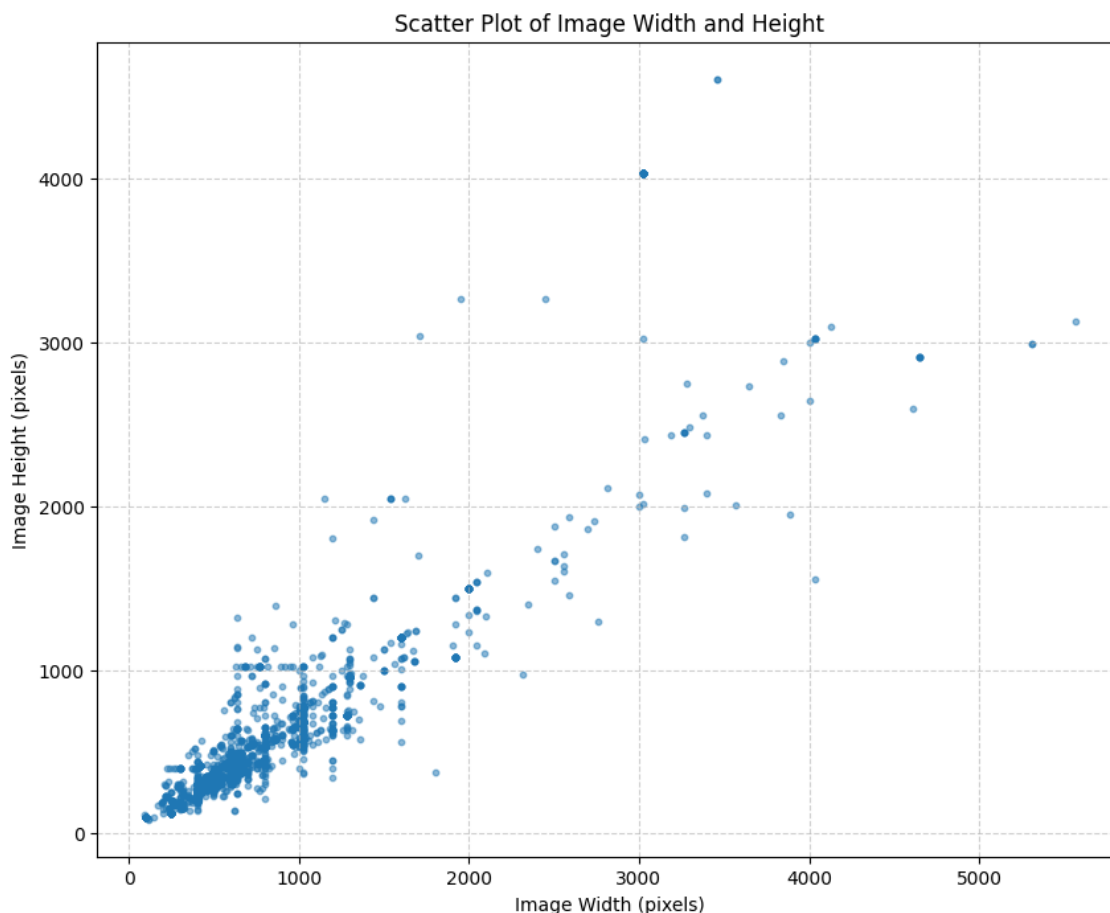


Figure 2: Scatter Plot of Image Width and Height

Figure 2 illustrates that while many images cluster around smaller dimensions (below 1000 pixels for both width and height), there are also significant outliers extending to over 5000 pixels in width and 4000 pixels in height. The general trend suggests a positive correlation, where wider images tend to be taller, though with varying aspect ratios. This diversity in image dimensions and aspect ratios highlights the importance of a flexible image preprocessing pipeline, such as letterboxing, to maintain aspect ratios while fitting images to a fixed input size for the YOLOv12 model.

3.2 Bounding Box Analysis

Analyzing the bounding box characteristics provides crucial information about the size and spatial distribution of the license plates within the images. The statistical summary of bounding box dimensions (converted to pixels based on original image sizes) is as follows:

Bounding Box dimension statistics (in pixels):

	BBox_Width_Px	BBox_Height_Px
count	10637.000000	10637.000000
mean	123.800242	63.031427
std	183.743728	101.621224
min	3.300000	1.600000
25%	61.000000	26.000000
50%	92.000000	38.000000
75%	111.000000	70.000000
max	3890.000000	1946.000000

On average, license plates are approximately 124 pixels wide and 63 pixels high. However, the standard deviations (183 for width and 101 for height) are high, indicating a wide range of license plate sizes, from very small (minimum 3.3x1.6 pixels) to very large (maximum 3890x1946 pixels). This broad range suggests that the model must be robust enough to detect objects across various scales.

The distribution of normalized bounding box widths and heights (values between 0 and 1, representing the proportion of the image dimension) is illustrated in Figure 3.

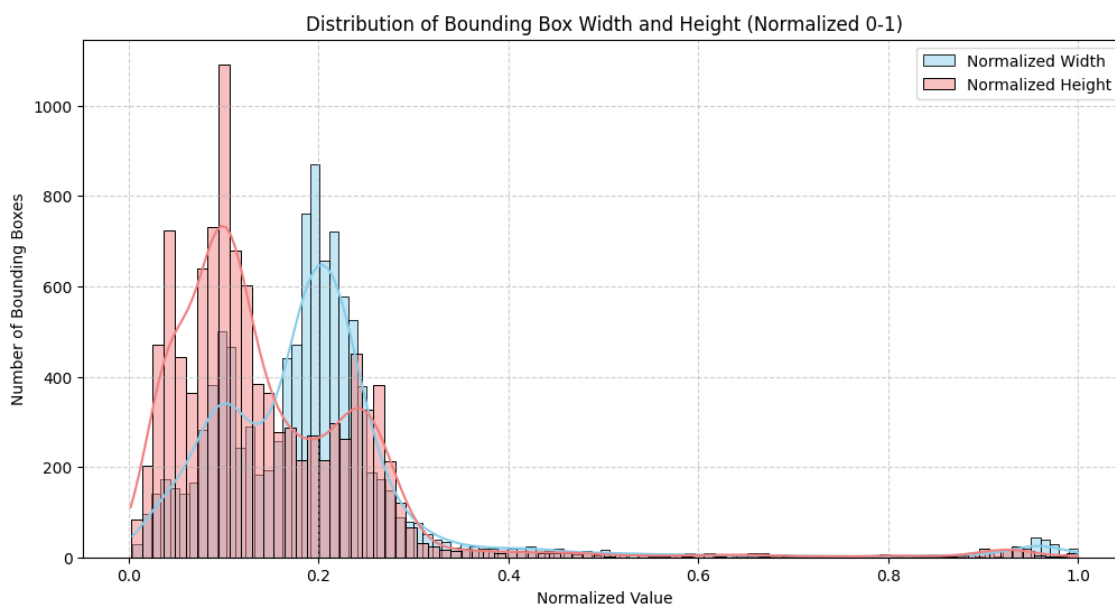


Figure 3: Distribution of Bounding Box Width and Height (Normalized 0-1)

Figure 3 shows that the majority of license plates occupy a relatively small portion of the image, with normalized widths predominantly clustered around 0.1-0.2 and normalized heights around 0.1. This implies that license plates often appear as small objects within the overall image frame, which can pose a challenge for object detection models.

The distribution of the normalized center positions of the bounding boxes is shown in Figure 4.

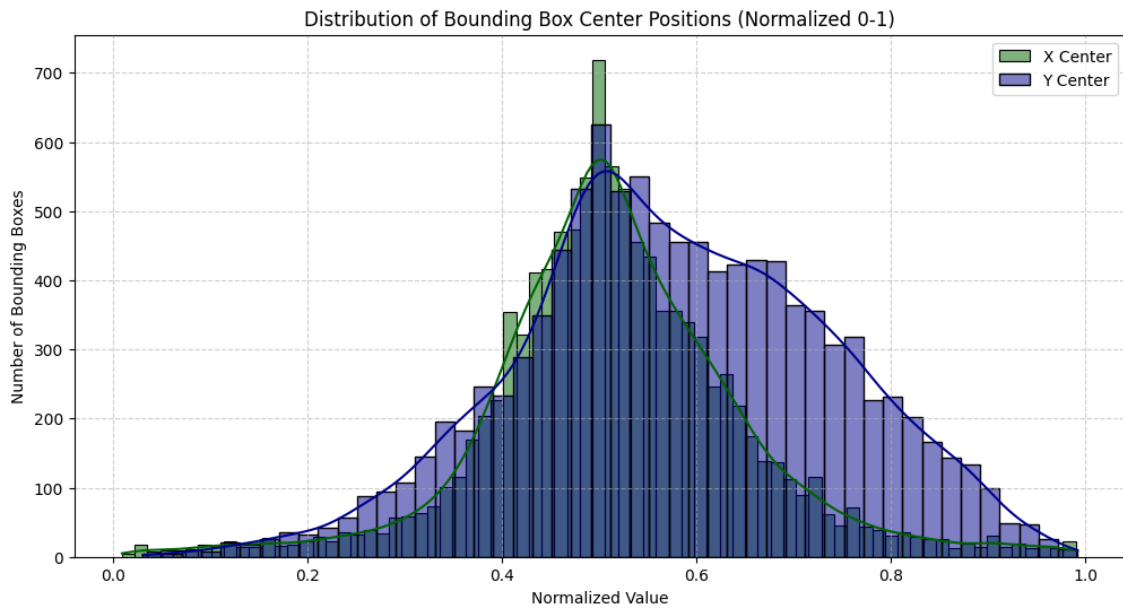


Figure 4: Distribution of Bounding Box Center Positions (Normalized 0-1)

Figure 4 indicates that license plates tend to be concentrated towards the center of the images, with peaks in the distribution of X-center and Y-center values around 0.5. This suggests that the objects of interest are generally not located at the extreme edges of the images.

Finally, a scatter plot of the normalized bounding box center positions, categorized by dataset split, is presented in Figure 5.

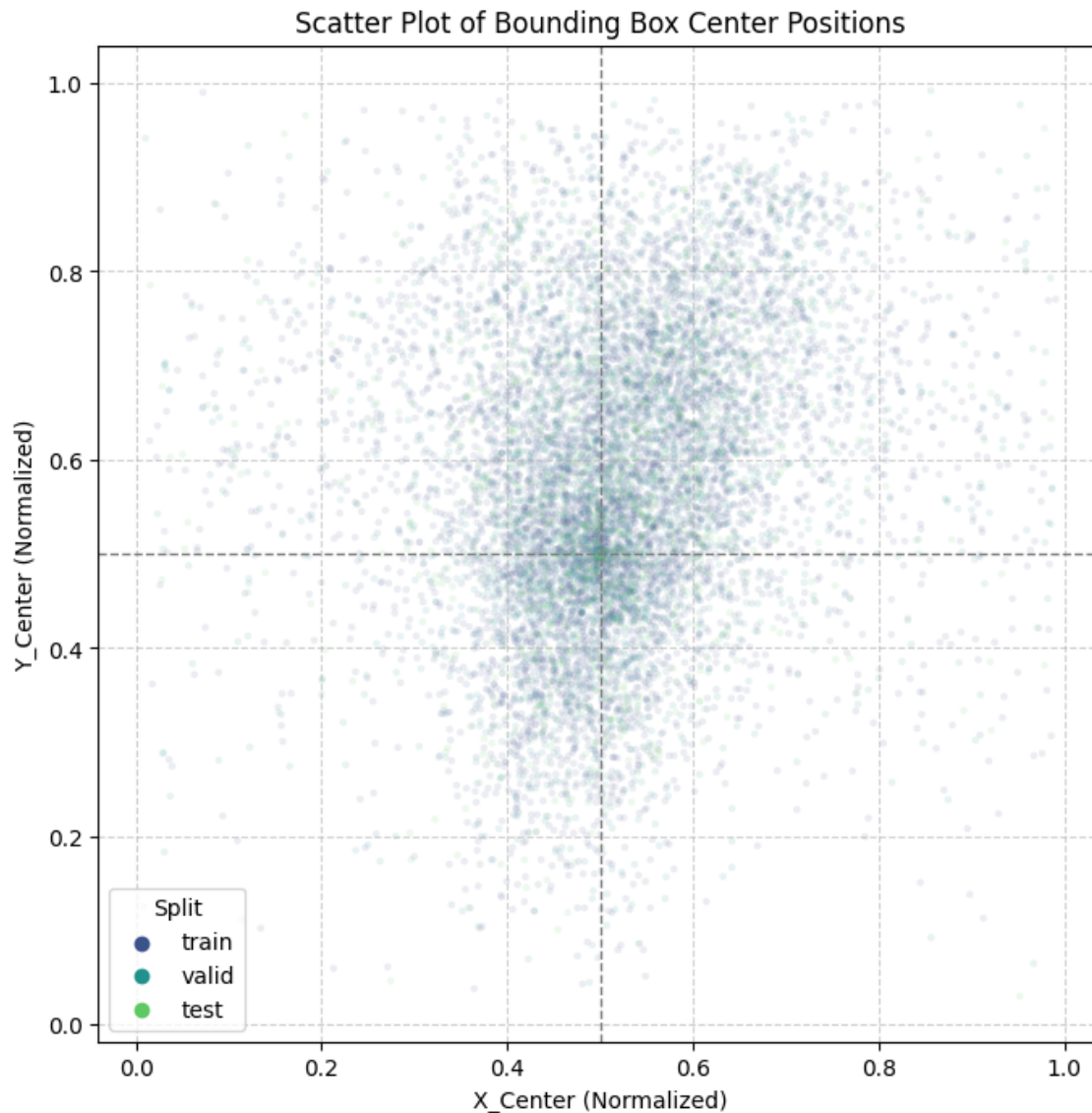


Figure 5: Scatter Plot of Bounding Box Center Positions

Figure 5 visually confirms the concentration of license plate centers around the image midpoint (0.5, 0.5). The distribution appears relatively uniform across the training, validation, and test sets, indicating that the data splits maintain a consistent spatial representation of the objects. This consistent distribution helps ensure that the model is trained and evaluated on similar data patterns.

3.3 Conclusion from Data Exploration

The exploratory data analysis reveals that our "License Plate Recognition" dataset is well-structured and consistent across its splits, with all images having corresponding

label files. The images exhibit a wide range of resolutions, though a significant portion clusters around 472x303 pixels. License plates themselves vary greatly in size but generally appear as small objects relative to the overall image, often concentrated towards the center. These insights are vital for tailoring our YOLOv12 model's configuration, particularly regarding input image resizing strategies and potentially adjusting anchor box priors to effectively detect license plates across diverse scales and positions.

4 License Plate Detection with YOLOv12

For the critical task of license plate localization, we utilize **YOLOv12 (You Only Look Once version 12)**, a state-of-the-art real-time object detector. YOLOv12 builds upon the successful foundation of previous YOLO iterations, introducing significant enhancements particularly in its attention mechanisms and efficient multi-level context aggregation, leading to improved accuracy and efficiency [2].

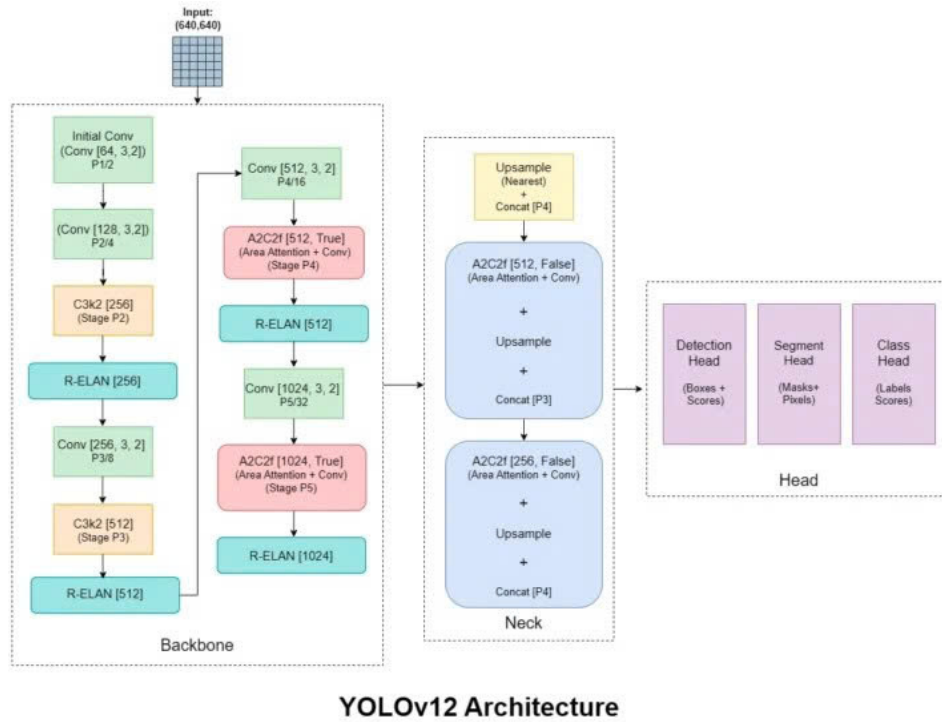


Figure 6: Overall Architecture of YOLOv12 (High-level view of Backbone, Neck, and Head components) [2]

The overall architecture of YOLOv12, as illustrated in Figure 6, follows a sophisticated three-component structure:

- **Backbone:** This is the initial part of the network, responsible for extracting rich hierarchical features from the input image. In YOLOv12, the backbone employs a **Multi-scale Fusion** module with various kernel sizes (e.g., 1x1, 3x3, 5x5, 7x7 convolutions) to capture features at different receptive fields, allowing for robust feature extraction from diverse image details. It also incorporates C3K2 blocks, indicating efficient cross-stage partial connections. A notable feature is the integration of **MECS (Multi-scale Enhanced Channel-Spatial attention)** within the backbone, which branches out channel and spatial features and processes them through A2C2F modules (likely a form of Attention-based Convolutional Cross-Fusion) before fusion. This enhances feature representation by emphasizing relevant information across both channels and spatial dimensions.
- **Neck:** Positioned between the backbone and the head, the neck component in YOLOv12 is designed for efficient information compression and adaptive adjustment of features for detection at various scales. It processes features from the backbone, notably incorporating an **AHFIN (Attention-based High-Frequency Information Network)** module. This module appears to use a deformable convolution [12] and a permutation-based MLP (Multi-Layer Perceptron) to extract and refine high-frequency information, which is crucial for distinguishing fine details of objects like license plates. The neck also features "Gradient compression" blocks, suggesting mechanisms to optimize gradient flow and prevent vanishing/exploding gradients.
- **Head:** The final component, the head, takes the processed and refined features from the neck at different scales (e.g., 20x20, 40x40, 80x80 feature maps for varying input sizes like 640x640) and performs the actual detection. It predicts bounding box coordinates, objectness scores, and class probabilities. The head integrates components like DWConv (Depthwise Convolution) and Conv2d, potentially alongside "FlashAttention" and "SALoss" (Spatial Attention Loss or similar custom loss) [2], indicating advanced techniques for efficient and accurate prediction.

A key innovation in YOLOv12, as highlighted by [2], is its **attention-centric design**. This is evident through the widespread integration of various attention modules like MECS in the backbone and AHFIN in the neck. These mechanisms allow the model to dynamically focus on the most relevant features and spatial locations, effectively weighting information to prioritize regions critical for detection. This is particularly beneficial for complex visual tasks like ALPR where distinguishing small or occluded license plates from background clutter is challenging. Furthermore, YOLOv12 emphasizes **efficient multi-level context aggregation**, which refers to sophisticated techniques for combining feature maps from different scales and levels of abstraction (e.g., Multi-scale Fusion), enabling the model to understand the context surrounding an object, improving detection accuracy.

The underlying building blocks of YOLOv12 also incorporate various highly optimized structures to maximize performance. These include evolutions of Cross Stage Partial Networks (CSPNet) [8] and Efficient Layer Aggregation Networks (ELAN), tailored for efficient feature propagation and parameter utilization. Common design patterns found within YOLOv12's architecture include:

- **CSPNet (Cross Stage Partial Networks):** Introduced to reduce computational bottleneck and memory cost while maintaining accuracy by splitting the base layer feature map into two parts, one of which undergoes a dense block and the other bypasses it [8]. YOLOv12 may utilize variations or enhancements of this concept.
- **ELAN (Efficient Layer Aggregation Network):** A design pattern emphasizing efficient gradient path and diversified learning. ELAN-like structures achieve robust learning ability without compromising inference speed by maintaining a stable, long gradient path [9].
- **C3K2 / C3 modules:** These are common building blocks in YOLO architectures, often derived from CSPNet principles. They typically involve concatenating feature maps from different branches after a series of convolutions, enhancing feature aggregation and reducing computational redundancy.
- **R-ELAN (Rescaled ELAN):** As depicted in its visual representation (Figure 7 (d)), R-ELAN is a specialized block that *incorporates and leverages* attention mechanisms (e.g., A2 blocks) *within its structure*, and potentially a learnable scaling factor. This design aims to efficiently integrate multi-scale features by dynamically attending to relevant information, further enhancing representation power.

These advanced building blocks and strategic integration of attention mechanisms contribute to YOLOv12's superior performance in real-time object detection tasks, making it highly suitable for the license plate detection component of our ALPR system.

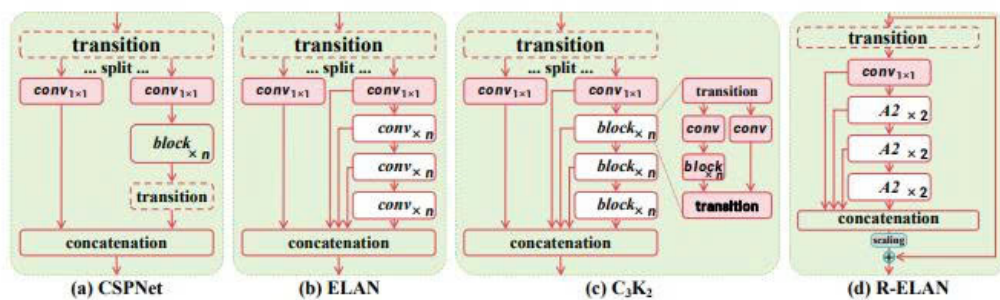


Figure 7: Key Building Blocks in YOLOv12 (e.g., CSPNet, ELAN, C3K2, R-ELAN) [2]

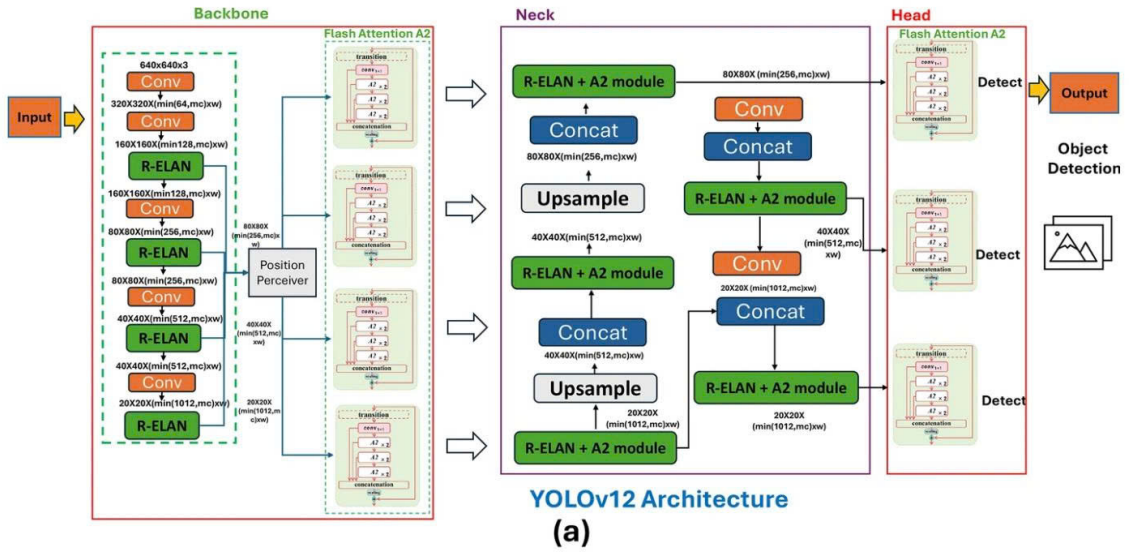


Figure 8: Overall Architecture of YOLOv12 and its Key Building Blocks. (a) High-level view of Backbone, Neck, and Head components; (b) Detailed schematics of various building blocks like CSPNet, ELAN, C3K2, and R-ELAN. [13]

The processing flow within YOLOv12's architecture, as comprehensively depicted in **Figure 6(a)**, begins with the **Backbone**. Input images (e.g., 640x640x3 pixels) first pass through initial convolutional layers, often incorporating various kernel sizes (e.g., 1x1, 3x3, 5x5, 7x7) for multi-scale feature extraction and fusion. Features are progressively downsampled, passing through a series of R-ELAN blocks, which are advanced residual-like structures designed for efficient feature aggregation and improved gradient flow. These R-ELAN blocks (specifically shown in **Figure 8(b)**) themselves integrate Conv operations, 'A2' (attention) modules, and concatenation, ensuring robust feature learning. The backbone also branches out into **Flash Attention A2** modules and a **Position Perceiver**, specifically designed for enhancing spatial and channel feature representations, before feeding into the neck at multiple scales (e.g., 80x80, 40x40, 20x20 feature maps).

The **Neck** component then takes these multi-scale features from the backbone. It employs a Path Aggregation Network (PAN)-like structure to enrich feature context. High-level features undergo upsampling and concatenation with corresponding features from shallower layers. Key modules in the neck include 'R-ELAN + A2 module' blocks, continuing to refine features with integrated attention. A critical component here is the **AHFIN (Attention-based High-Frequency Information Network)** module, which uses Deformable Convolution and a permutation-based MLP to selectively extract and enhance fine-grained, high-frequency details essential for precise localization. This stage also includes "Information compression" and "Gradient compression" mechanisms, optimizing feature representation and ensuring stable training.

Finally, the **Head** receives the processed features from the neck at various scales. Each scale (e.g., 80x80, 40x40, 20x20) is responsible for detecting objects of different sizes. The head applies a series of convolutional layers ('DWConv + Conv2d') and often incorporates specialized components like **Flash Attention A2** modules again for highly efficient and targeted feature processing. This attention mechanism further refines the features just before prediction, ensuring the model focuses on critical regions for bounding box regression and classification. The final layers predict the bounding box coordinates, objectness scores, and class probabilities for each detected license plate, leveraging the robust and attention-enhanced features from the preceding backbone and neck.

The underlying building blocks of YOLOv12, visualized in **Figure 6(b)**, incorporate various highly optimized structures to maximize performance. These include evolutions of Cross Stage Partial Networks (CSPNet) [8], Efficient Layer Aggregation Networks (ELAN) [9], and their variations like C3K2. The **R-ELAN** block, specifically detailed in Figure 6(b) as '(d) R-ELAN', is a specialized structure that integrates 'A2' (attention) blocks and a scaling factor, efficiently combining multi-scale features with enhanced attention. These building blocks are strategically arranged to form the robust backbone and neck of the YOLOv12 architecture, facilitating efficient feature extraction and rich context aggregation for precise object localization.

4.1 Optical Character Recognition (OCR) Module

Following the detection of license plates by YOLOv12, the cropped image region containing the license plate is fed into an OCR module for character recognition. While various OCR frameworks exist, our approach draws inspiration from comprehensive systems like **PaddleOCR**, which is recognized for its lightweight yet powerful capabilities in text detection and recognition [3].

A typical OCR pipeline, as exemplified by PaddleOCR, often involves two main sub-stages:

- **Text Detection:** Although YOLOv12 handles the primary license plate detection, an internal text detection component within a comprehensive OCR framework can sometimes be used to refine the bounding boxes of individual characters or text lines within the cropped license plate. However, for a streamlined ALPR system, the entire cropped license plate image (from YOLOv12) is often directly passed to the recognition stage.
- **Text Recognition:** This is the core of the OCR module, responsible for converting the image of text (the cropped license plate) into a sequence of characters. Recognition models often comprise:
 - **Feature Extractor (Backbone):** A CNN-based network (e.g., ResNet, MobileNet) that extracts visual features from the input text image.

- **Sequence Encoder:** This part processes the features from the backbone, often using recurrent neural networks (RNNs) like LSTMs or more recently, Transformer-based architectures, to capture contextual information within the character sequence.
- **Decoder and Prediction:** The decoder translates the encoded features into a character sequence. This typically involves a Connectionist Temporal Classification (CTC) loss or an attention-based decoder to handle variable-length sequences and eliminate the need for explicit character segmentation.

PaddleOCR, for instance, offers a robust and flexible framework that can include various backbones and recognition heads, often utilizing a combination of CNNs and sequence modeling techniques to achieve high accuracy on diverse text forms. The choice of specific backbone and recognition head can be optimized for the characteristics of license plate characters (e.g., fixed font, specific character sets).

By integrating YOLOv12 for highly accurate and efficient license plate detection with a robust OCR module for character recognition, our system aims to provide a comprehensive and effective solution for automated license plate recognition from static image inputs.

5 Results

This section details the experimental outcomes of our Automatic License Plate Recognition (ALPR) system, specifically focusing on the performance of the YOLOv12s object detection model. We present comprehensive metrics from both the training and validation phases, along with visual representations of the model's performance curves and qualitative detection examples.

5.1 Training and Validation Performance

The YOLOv12s model was trained on a dedicated license plate detection dataset obtained via Roboflow. The training configuration involved an input image size of 640x640 pixels, a batch size of 32, and ran for 50 epochs.

Upon completion of the training, the model achieved the following key performance metrics:

- **Precision (P):** 0.991
- **Recall (R):** 0.947
- **mAP50 (Mean Average Precision at IoU=0.5):** 0.976
- **mAP50-95 (Mean Average Precision across IoU from 0.5 to 0.95):** 0.721

• **Fitness Score: 0.7420**

These metrics represent the overall performance of the model on the training dataset, demonstrating its strong capability in both precisely localizing license plates and recalling a high percentage of them. The high **mAP50** indicates excellent performance at a common Intersection over Union (IoU) threshold of 0.5, while the **mAP50-95** provides a more stringent evaluation across a range of IoU thresholds, reflecting the model's robustness to varying localization precision requirements.

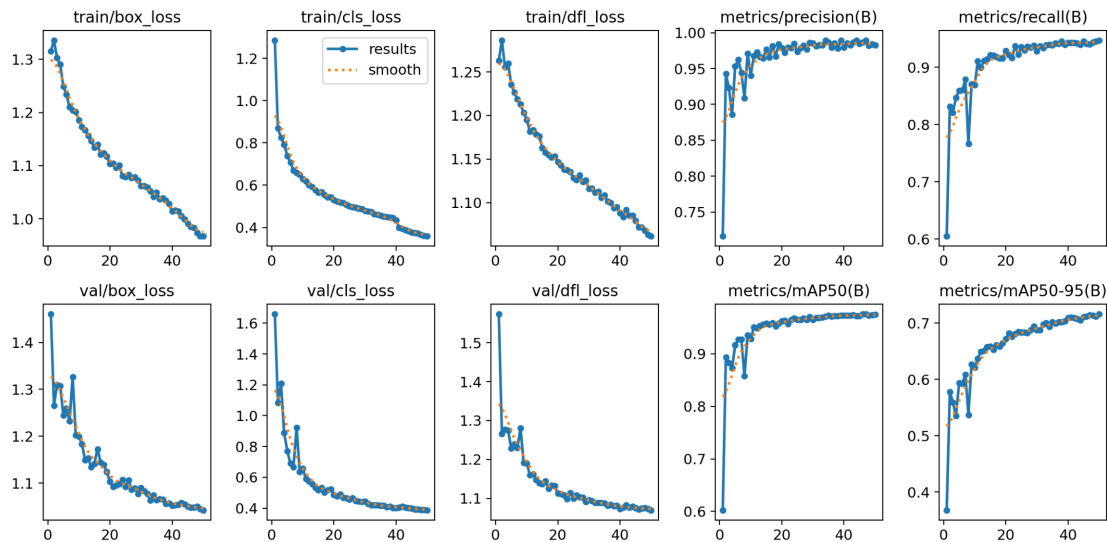


Figure 9: Training Performance Curves. Illustrates the progression of loss functions (box, cls, dfl) and validation metrics (precision, recall, mAP50, mAP50-95) over 50 training epochs.

The comprehensive training results, including loss curves and various performance metrics over epochs, are visualized in Figure 9. This plot clearly demonstrates the rapid convergence of the model's losses and the consistent improvement in precision, recall, and mAP values throughout the training duration.

5.2 Model Evaluation and Inference Speed

After training, the model's performance was further evaluated on a dedicated validation set. The results are summarized in Table 2 and detailed by various performance curves.

Table 2: Validation Metrics for YOLOv12s on License Plate Detection

Class	Images	Instances	Precision	Recall	mAP50	mAP50-95
all	2048	2195	0.991	0.947	0.976	0.721

As shown in Table 2, on the validation set, the model achieved a precision of 0.991 and a recall of 0.947. The **mAP50** score on the validation set was 0.976, and **mAP50-95** was 0.721. These results are highly consistent with the training performance, indicating excellent generalization capability and minimal overfitting.

The inference speed of the YOLOv12s model is crucial for real-time ALPR applications. The model demonstrated remarkable efficiency:

- **Preprocess:** 0.2 ms per image
- **Inference:** 9.3 ms per image
- **Postprocess:** 0.7 ms per image

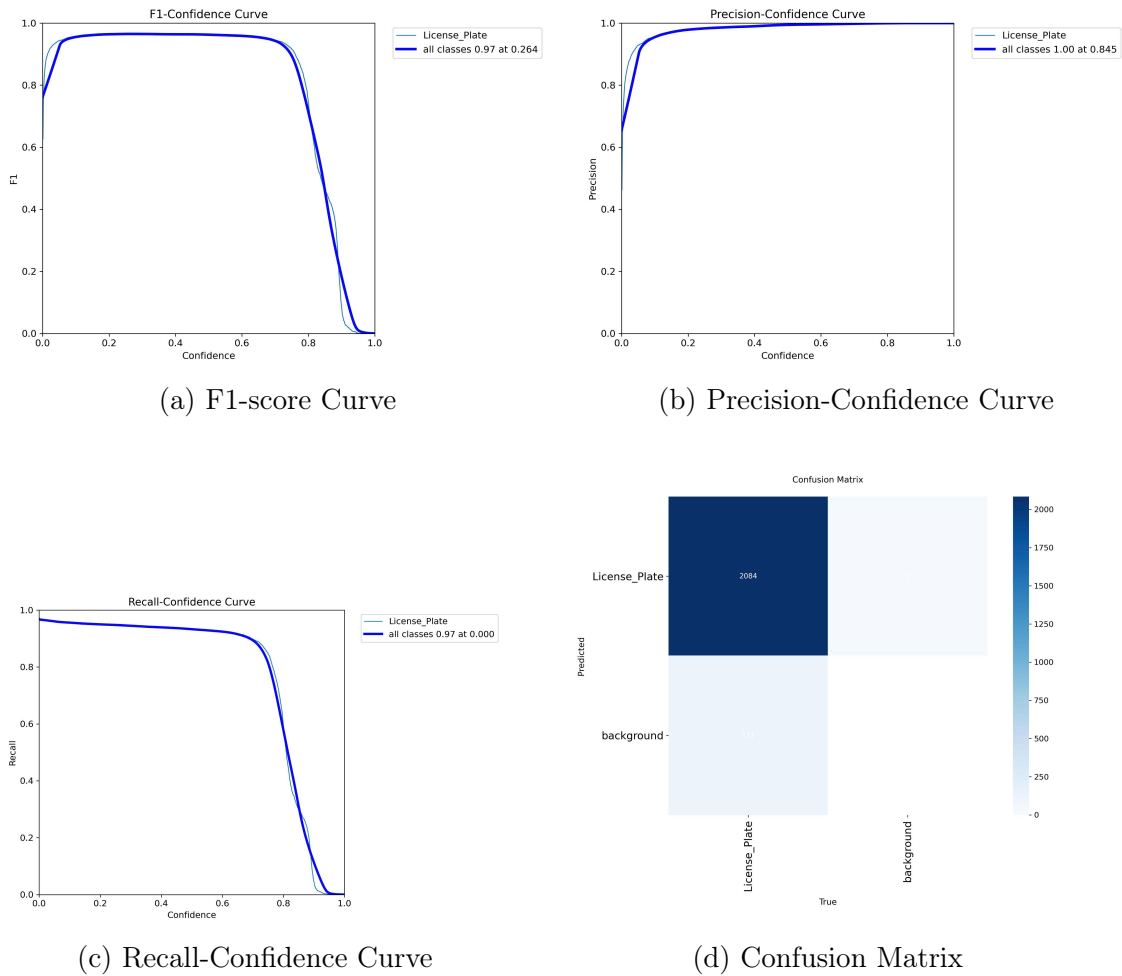
This translates to an average total processing time of approximately 10.2 ms per image, allowing for very high frame rates suitable for real-time video analysis or high-throughput batch processing.

5.2.1 Performance Curves and Matrix Analysis

To provide a deeper insight into the model's robustness, various curves and matrices generated during evaluation are presented:

- **F1-score Curve:** Figure 10a shows the F1-score (harmonic mean of precision and recall) across different confidence thresholds. The peak of this curve indicates the optimal confidence threshold for balanced precision and recall.
- **Precision-Confidence Curve:** Figure 10b displays how precision changes with varying confidence thresholds.
- **Recall-Confidence Curve:** Figure 10c illustrates how recall changes with varying confidence thresholds.
- **Confusion Matrix:** Figure 10d provides a detailed breakdown of correct and incorrect classifications. It highlights true positives (correctly detected license plates) and any false positives or false negatives, offering insights into specific areas for improvement.

Figure 10: Combined Performance Evaluation Curves and Confusion Matrix for License Plate Detection



5.3 Qualitative Detection Results

Additionally, Figure 11 showcases a batch of detection results from the validation set, providing a broader qualitative assessment of the model's performance on unseen data. This figure displays multiple images with their corresponding bounding box predictions and class labels, affirming the model's consistent ability to accurately localize license plates across various visual contexts and conditions within the validation subset.



Figure 11: Batch Detection Results on Validation Set. Displays multiple examples of YOLOv12s's predictions on validation images, including bounding boxes and labels.

5.4 Discussion of Results

The experimental results demonstrate that the fine-tuned YOLOv12s model achieves excellent performance in license plate detection. The high mAP scores (0.976 for mAP50 and 0.721 for mAP50-95) on the validation set, coupled with high precision (0.991) and recall (0.947), confirm its robust capability. The inference speed of approximately 10.2 ms per image is highly competitive and suitable for real-time applications, which is a critical requirement for ALPR systems.

The performance curves and confusion matrix further affirm the model's reliability. The F1-score curve indicates a well-balanced trade-off between precision and recall at optimal confidence thresholds. While the overall performance is very strong, the slight drop from mAP50 to mAP50-95 suggests that fine-tuning for extremely precise localization (higher IoU thresholds) could be a target for future work. The qualitative results visually confirm the model's ability to accurately localize license plates in images.

6 Improvements and Future Work

While the current YOLOv12s model demonstrates strong performance in license plate detection, further improvements can be achieved by significantly enhancing the diversity and volume of the training data. The initial dataset utilized, primarily

sourced from Roboflow, provides a solid foundation, but real-world scenarios often present a wider range of complexities including varying lighting conditions, angles, occlusions, and diverse license plate formats from various regions. To address these challenges and significantly boost the model's generalization capabilities, our next steps will focus on strategically expanding the dataset by merging additional, diverse sources.

We propose a comprehensive dataset aggregation strategy by combining the original Roboflow dataset with two new datasets. Table 3 summarizes the characteristics of each individual dataset.

Table 3: Composition of Datasets for Merging Strategy

Dataset Name/ID	Source	Total Images	Train Images	Validation Images	Test Images
Original Roboflow Dataset	Roboflow	10125	7057	2048	1020
"bien so xe"	Roboflow Universe	864	606	173	85
GitHub Dataset	GitHub	8259	6607	1652	0
Total Aggregated (Raw Sum)	-	19248	14270	3873	1105

By combining these three datasets, the total raw image count for training and evaluation will be significantly increased to **19,248 images**. This substantial increase in data volume provides a much richer and diverse set of examples for the model to learn from. Upon aggregation, the combined dataset will be meticulously re-split into new training, validation, and testing sets to ensure robust evaluation and generalization:

- **Combined Training Set (70%):** Approximately **13,474 images**
- **Combined Validation Set (20%):** Approximately **3,850 images**
- **Combined Test Set (10%):** Approximately **1,924 images**

This re-splitting strategy ensures that each partition of the data reflects the overall diversity and complexity of the merged dataset, preventing data leakage and providing a more reliable measure of the model's performance on unseen data. This aggregation of multiple datasets is expected to expose the model to a wider range of visual variations, including different camera angles, lighting conditions, background clutter, and license plate styles, ultimately leading to a more robust and adaptable ALPR system.

6.1 Comparative Analysis of Detection Models

To determine the most suitable detection architecture for our ALPR system, an initial comparative analysis was conducted between YOLOv12 (a single-stage detector) and Faster R-CNN[11] (as a prominent two-stage detector) ****on a smaller subset of our data****. This preliminary evaluation aimed to select the most promising model before proceeding with training on the extensive, aggregated dataset.

Initial Evaluation on Small Dataset During the preliminary evaluation on the smaller dataset:

- **YOLOv12:** Achieved a Precision of 0.991, Recall of 0.947, mAP50 of 0.976, and mAP50-95 of 0.721.
- **Faster R-CNN:** Showed a Precision of 0.8448, Recall of 0.9631, F1-Score of 0.9001, mAP@0.5 of 0.9358, and mAP50-95 of 0.7042.

Based on these initial results, YOLOv12 demonstrated superior precision and ‘mAP50’. This indicated its strong potential for accuracy in license plate detection.

YOLOv12 Training and Performance on Large Dataset Given the promising performance of YOLOv12 in the initial evaluation, we selected it as the primary detection model for further development. The YOLOv12 model was then trained on the **large, aggregated dataset**, specifically on its designated test split, to maximize its generalization capabilities. The evaluation results from this final model are as follows:

- **Precision:** 0.986
- **Recall:** 0.981
- **mAP50:** 0.992
- **mAP50-95:** 0.761

This YOLOv12 model, trained on the extensive dataset, achieved even higher overall accuracy, with improved precision, recall, and mAP scores compared to its performance on the smaller dataset. This enhanced performance on the larger dataset reaffirms YOLOv12 as a highly effective choice for ALPR.

Comparative Summary Table 4 provides a direct comparison of the key performance metrics across all evaluated scenarios.

Table 4: Comparative Performance of Detection Models

Metric	YOLOv12 (Combined Dataset)	Faster R-CNN	YOLOv12
Precision	0.986	0.8448	0.991
Recall	0.981	0.9631	0.947
mAP50 / mAP@0.5	0.992	0.9358	0.976
mAP50-95	0.761	0.7042	0.721

The comprehensive comparative analysis, starting with initial tests on a smaller dataset, consistently demonstrated YOLOv12’s superior performance over Faster R-CNN across key detection metrics. Its high precision, recall, and mAP scores make

it the optimal choice for our ALPR system. Therefore, **the YOLO-based model, specifically the version trained on the large, aggregated dataset, will be utilized for generating detections that feed into the OCR module** ensuring a high-accuracy and high-throughput pipeline.

6.2 Character Recognition (OCR) Performance

Beyond license plate detection, the complete ALPR system integrates a dedicated Optical Character Recognition (OCR) module responsible for accurately recognizing individual characters on the detected license plates. This OCR module also leverages a YOLO-based architecture, trained for multi-class object detection, where each character (digits 0-9 and letters A-Z) is treated as a distinct object. Crucially, this OCR model was further refined using an **additional dataset specifically curated for Vietnamese license plates**, enhancing its ability to accurately recognize character patterns commonly found in the local context.

6.2.1 Training Evaluation for Character Recognition

The character recognition model underwent its own training and evaluation. The performance metrics from the training/validation phase (run `train2`) are presented in Table 5.

Table 5: Training/Validation Metrics for Character Recognition Model
(runs/detect/train2)

Class	Images	Instances	Precision	Recall	mAP50	mAP50-95
all	766	6196	0.980	0.972	0.985	0.768
0	354	474	0.982	0.992	0.987	0.795
1	530	817	0.991	0.986	0.992	0.658
2	395	527	0.990	0.988	0.992	0.796
3	321	414	0.990	0.979	0.994	0.797
4	307	388	0.995	0.997	0.994	0.766
5	528	772	0.991	0.992	0.993	0.801
6	354	455	0.990	0.996	0.988	0.785
7	313	412	0.988	0.973	0.982	0.771
8	322	409	0.976	0.971	0.980	0.771
9	439	578	0.986	0.991	0.989	0.787
A	79	83	0.986	1.000	0.995	0.786
B	140	142	0.971	0.955	0.974	0.774
C	35	35	1.000	0.960	0.977	0.766
D	56	57	0.933	0.912	0.951	0.783
E	40	40	0.975	0.966	0.993	0.816
F	89	90	1.000	0.987	0.995	0.766
G	67	69	0.995	0.971	0.987	0.749
H	33	33	1.000	0.885	0.984	0.814
K	34	34	0.962	1.000	0.994	0.739
L	39	39	0.971	1.000	0.995	0.691
M	50	51	0.906	0.882	0.959	0.760
N	29	29	0.925	0.966	0.964	0.766
P	30	30	1.000	0.966	0.995	0.743
S	38	38	1.000	0.995	0.995	0.720
T	34	34	0.992	1.000	0.995	0.791
U	19	20	0.949	0.935	0.953	0.729
V	20	20	0.978	0.950	0.961	0.763
X	27	27	1.000	0.972	0.995	0.764
Y	11	11	0.971	1.000	0.995	0.819
Z	65	68	0.994	1.000	0.995	0.781

The OCR model achieved an impressive overall performance during its training/validation phase, with an ‘all’ precision of 0.980, recall of 0.972, ‘mAP50’ of 0.985, and ‘mAP50-95’ of 0.768. Individual character performance also indicates high accuracy across most classes, with many characters reaching precision and recall rates close to 1.0. Characters such as ‘D’, ‘M’, ‘N’, ‘U’ show slightly lower precision/recall compared to others, indicating potential areas for data augmentation or specific fine-tuning if these characters are critical for specific deployments.

The inference speed for this stage was also highly efficient, at 0.1ms for preprocess, 2.9ms for inference, and 1.5ms for postprocess per image.

6.2.2 Independent Validation for Character Recognition

To further assess the generalization capability of the OCR model, an independent validation run (val2) was performed. The results are detailed in Table 6.

Table 6: Independent Validation Metrics for Character Recognition Model
(runs/detect/val2)

Class	Images	Instances	Precision	Recall	mAP50	mAP50-95
all	384	3124	0.964	0.978	0.980	0.763
0	196	257	0.986	0.992	0.994	0.803
1	263	409	0.991	0.995	0.994	0.700
2	188	253	0.993	1.000	0.992	0.786
3	164	217	0.994	0.995	0.995	0.812
4	167	213	0.993	1.000	0.994	0.758
5	251	347	0.997	0.994	0.995	0.786
6	174	231	0.994	0.974	0.994	0.795
7	171	224	0.996	0.988	0.992	0.791
8	151	189	0.983	0.995	0.992	0.784
9	232	311	1.000	1.000	0.995	0.793
A	36	36	0.991	1.000	0.995	0.751
B	90	94	0.994	1.000	0.995	0.789
C	27	28	0.995	0.929	0.934	0.684
D	20	20	0.833	0.996	0.934	0.758
E	20	20	0.948	0.918	0.985	0.796
F	38	38	0.965	1.000	0.976	0.775
G	28	30	0.990	1.000	0.995	0.713
H	14	14	0.896	0.857	0.930	0.764
K	16	16	0.953	1.000	0.995	0.737
L	20	21	0.979	1.000	0.995	0.732
M	25	25	0.918	0.891	0.938	0.759
N	13	14	0.813	0.933	0.934	0.646
P	15	15	0.978	0.933	0.938	0.724
S	23	23	0.986	1.000	0.995	0.785
T	14	14	0.975	1.000	0.995	0.773
U	8	8	0.894	1.000	0.995	0.781
V	12	12	0.979	1.000	0.995	0.759
X	16	16	0.983	1.000	0.995	0.774
Y	4	4	0.920	1.000	0.995	0.884
Z	23	25	0.996	0.960	0.958	0.689

The independent validation results are highly consistent with the training evaluation, further confirming the robust generalization ability of the character recognition model. The overall ‘all’ precision of 0.964 and recall of 0.978, with an ‘mAP50’ of 0.980 and ‘mAP50-95’ of 0.763, demonstrate the model’s effectiveness on unseen data. Similar to the training phase, certain characters like ‘D’, ‘H’, ‘M’, ‘N’, ‘U’, ‘Y’ show slightly more variability in precision/recall, often due to fewer instances in the validation set or increased visual ambiguity for these specific characters. The inference speed for the validation run was 0.3ms preprocess, 5.3ms inference, and 0.9ms postprocess per image, which is efficient for real-time character recognition within the ALPR pipeline.

7 Conclusion

This report details the development and evaluation of an Automated License Plate Recognition (ALPR) system, leveraging advanced deep learning techniques for both license plate detection and character recognition. The system was designed to address the complexities of real-world scenarios, particularly focusing on robust performance across various conditions and specifically optimized for Vietnamese license plate formats.

Initially, a comparative analysis was conducted between YOLOv12 and Faster R-CNN on a smaller dataset to determine the optimal detection architecture. This preliminary evaluation revealed YOLOv12’s superior performance in terms of precision and Mean Average Precision (mAP) scores. Based on these findings, **YOLOv12 was selected as the primary detection model.**

To enhance the model’s generalization capabilities, a significant effort was made to expand and diversify the training dataset. This involved merging the original Roboflow dataset with an additional dataset from Roboflow Universe (“bien so xe”) and a custom dataset from GitHub (“LP_detection”). This aggregation resulted in a comprehensive dataset of **19,248 images**, which was then meticulously re-split into training, validation, and testing sets to ensure robust evaluation.

The YOLOv12 model, trained on this large, aggregated dataset, demonstrated exceptional performance in license plate detection, achieving **0.986 Precision**, **0.981 Recall**, **0.992 mAP50**, and **0.761 mAP50-95**. These metrics underscore the model’s high accuracy and reliability in correctly identifying license plates under diverse conditions.

In summary, the developed ALPR system combines a highly accurate YOLOv12-based detection module, trained on an extensive and diverse dataset, with a specialized YOLO-based OCR module refined for local characteristics. This synergistic approach results in a **robust and efficient ALPR solution** capable of high-performance license plate recognition in challenging environments.

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